

Problem A:

1936: Alan Turing made the Turing machine

1947: Transistor invented at Bell Labs by John Bardeen, Walter Brattain, and William Shockley

1958: Invention of the microchip by Jack Kilby at Texas Instruments

1969: Launch of ARPANET, also known as the first message sent from UCLA to Stanford

1981: IBM released the IBM Personal Computer

1991: Tim Berners-Lee released the World Wide Web to the public

2000: Release of the Intel Pentium 4 processor

2007: Launch of the iPhone

2012: Release of the Raspberry Pi

2022: Release of ChatGPT by OpenAI

Problem C:

(a) Prediction for 2026

- Reference values: $T_0 = 2,300$, reference year = 1971.
- Elapsed years (t): $2026 - 1971 = 55$ years.
- Formula application: $T(55) = 2300 * 2^{(55 / 2)} = 4.3657 * 10^{11}$

Predicted Transistor Count: Approximately 437 billion transistors

(b) Comparison with 2026 Processors

Comparing:

Predicted(Moore's Law): Approximately 437 billion transistors

Actual(Apple M5, 2026): Approximately 28 billion transistors

The prediction overshoots the actual value by roughly 15.6 times. So while Moore's Law captured the general exponential trend remarkably well for decades, the strict 2-year doubling period has not held exactly over the full 55-year span.

(c) Calculating the Actual Doubling Period

$2300 * 2^{(55 / d)} = 28,000,000,000$ [Taking the actual value and using logarithm]

$\Rightarrow d = 2.34$ years

Actual Doubling Period: 2.34 years

Problem D:

(a) What's wrong with the pseudocode?

The bug is that the algorithm checks for a KO for each individual event, rather than checking for all events that share the same frame.